MOHITH DODAPANENI BALARAJU

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EDUCATION

University of Houston, Houston, TX Aug 2025 – May 2027
Master of Science in Engineering Data Science and AI GPA: 3.33/4.0

TECHNICAL SKILLS

Languages & Databases: SQL, Python, Snowflake, PostgreSQL, MySQL, Databricks, dbt

Data Engineering: ETL, ELT, Medallion Architecture (Bronze/Silver/Gold), Data Pipelines, Schema Validation, Data Quality Testing, Pandas, NumPy, Pydantic

Machine Learning: Scikit-learn, LightGBM, XGBoost, Random Forest, SHAP, Feature Engineering, Walk-Forward Validation, PyTorch, TensorFlow

LLM & Cloud Platforms: LLM, RAG, Prompt Engineering, FastAPI, Streamlit, Gemini,AWS,GCP

Analytics & Visualization: Pandas, Streamlit, Power BI, Excel, Jupyter Notebook

EXPERIENCE

Instructional Assistant (Teaching Assistant) — *University of Houston – Bauer College of Business* Aug 2025 – Present
Houston, TX

- Built an end-to-end data pipeline using Pandas and Excel to clean, restructure, and maintain demographic and academic records for 3,200+ students.
- Developed Power BI advising dashboards analyzing student geographic pipelines and admission distributions across a 3-month reporting cycle.
- Utilized AI developer assistants to generate, optimize, and troubleshoot complex DAX queries and data transformation logic, ensuring 100% metric accuracy.
- Guided 400+ students through technical course concepts, resolved curriculum doubts, and collaborated with faculty to develop modular lab materials and documentation.

PROJECTS

Type 2 Diabetes Trial-Screening Lakehouse GitHub
Technologies: Python, Snowflake, dbt, Streamlit, Gemini

- Landed Synthea patients, observations, conditions, and medications plus 50 recruiting ClinicalTrials.gov protocols in Snowflake bronze, then built dbt silver models for typed demographics, type 2 diabetes conditions, and latest HbA1c (LOINC 4548-4).
- Scored every patient–trial pair in a Gold fact (58,550 rows) as eligible, needs-review, or ineligible from age windows and a parsed inclusion-text HbA1c bound, with 29 dbt tests and direct identifiers kept in bronze.
- Built a Streamlit desk for the eligible, needs-review, and ineligible cohorts, with a Gemini copilot that answers from the protocol text and does not change match status.

ERCOT Real-Time Lakehouse and Price-Spike Forecaster GitHub
Technologies: Python, Snowflake, dbt, LightGBM, SHAP, FastAPI, Streamlit

- Landed 5-minute HB HOUSTON and HB WEST SCED telemetry (LMP, operating reserves, net load, and heat index) in Snowflake bronze, then built dbt silver features for reserve-depletion velocity, congestion spread, and net-load acceleration.
- Implemented walk-forward LightGBM training for a 60-minute-ahead spike label (LMP above $250/MWh), with an embargo so the current price cannot leak into the target, and served probabilities plus SHAP attributions from FastAPI.
- Built a Streamlit desk that compares taking the price, shedding up to 20 MW, and discharging a 10 MW battery for a 50 MW Houston Ship Channel load, including downtime cost and 4CP transmission exposure.

CERTIFICATIONS

- Preparing Data for Analysis with Microsoft Excel – Microsoft — Coursera
- Foundations: Data, Data, Everywhere – Google — Coursera
- Everything About Machine Learning – Duke University — Coursera
- The Ultimate MySQL Bootcamp: Go from SQL Beginner to Expert – Udemy
- AWS Certified Solutions Architect – Associate (2026) – Udemy

CURRENT INITIATIVES & UPCOMING MILESTONES

- **Certifications & Cloud Architecture:** Actively preparing for the **AWS Certified Data Engineer – Associate** credential to deepen expertise in distributed data ingestion and cloud pipeline optimization.
- **Academic Research:** Collaborating on clinical data modeling for short-term mortality risk assessment using MIMIC-IV datasets under faculty mentorship.